Deliverable 5: Normalization

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

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Team Information

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1 Task 1: Lab 4 Schema BCNF Analysis

1. All relations in the Lab 4 schema are already in BCNF. None of the relations violate BCNF.

Justification: For every non-trivial functional dependency $X \rightarrow Y$, the determinant X is a superkey.

2 Task 2: Normalization Practice

a) Equipment Inventory

- **Candidate Key:** `serialNo`
- **BCNF:** No. The dependency `modelId` → `manufacturer` violates BCNF because `modelId` is not a superkey.
- **Strongest NF:** 1NF (since it is not in 3NF).
- **Decomposition:** R1(modelId, manufacturer) and R2(serialNo, modelId, labId).
- **Properties:** The decomposition is lossless and preserves all functional dependencies.

b) Project Participation

- **Candidate Key:** `(pid, projectCode)`
- **BCNF:** No. The dependency `projectCode` → `piPid` violates BCNF because `projectCode` is not a superkey.
- **Strongest NF:** 1NF.
- **Decomposition:** R1(projectCode, piPid) and R2(pid, projectCode, role).
- **Properties:** Lossless and dependency-preserving.

c) Equipment Certification Requirement

- **Candidate Key:** `(modelId, certCode)`
- **BCNF:** No. The dependency `certCode` → {`certTitle, issuing Authority`} violates BCNF because `certCode` is not a superkey.
- **Strongest NF:** 1NF.
- **Decomposition:** R1(certCode, certTitle, issuing Authority) and R2(modelId, certCode).
- **Properties:** Lossless and dependency-preserving.

d) Supplier Offer

- **Candidate Key:** `(suppId, consId, labId)`
- **BCNF:** No. The dependency `suppId` → `supplier Name` violates BCNF because `suppId` is not a superkey.
- **Strongest NF:** 1NF.
- **Decomposition:** R1(suppId, supplier Name) and R2(suppId, consId, labId, unitPrice).
- **Properties:** Lossless and dependency-preserving.

e) Reservation and Project Information

- **Candidate Key:** `resId`
- **BCNF:** No. The dependency `projectCode` → `projectTitle` violates BCNF because `projectCode` is not a superkey.
- **Strongest NF:** 1NF.
- **Decomposition:** R1(projectCode, projectTitle) and R2(resId, projectCode, madeByPid, status).
- **Properties:** Lossless and dependency-preserving.